Lab Book

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Preface

This book details my research during Summer 2026. Many thanks to Dr. Andrew Young and Mr Darius Michienzi for supervising this project.

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1 Reading

1.1 Summary of "Broad Iron Lines in Active Galactic Nuclei" [1]

Fluorescent iron line, Fe-Kα, in the X-ray band at $6.4 - 6.9$ keV is the strongest narrow line emitted by an accretion disk around a black hole. Inverse Compton scattering of the soft photons (lower energy than the reactant particles) in a hot corona is the most likely candidate for the production of these X-rays. The emitted line is broadened and skewed due to the Doppler effect and gravitational redshift, and is often called the broad iron line due to being broadened to a FWHM of 5×10^4 km s^{-1}.

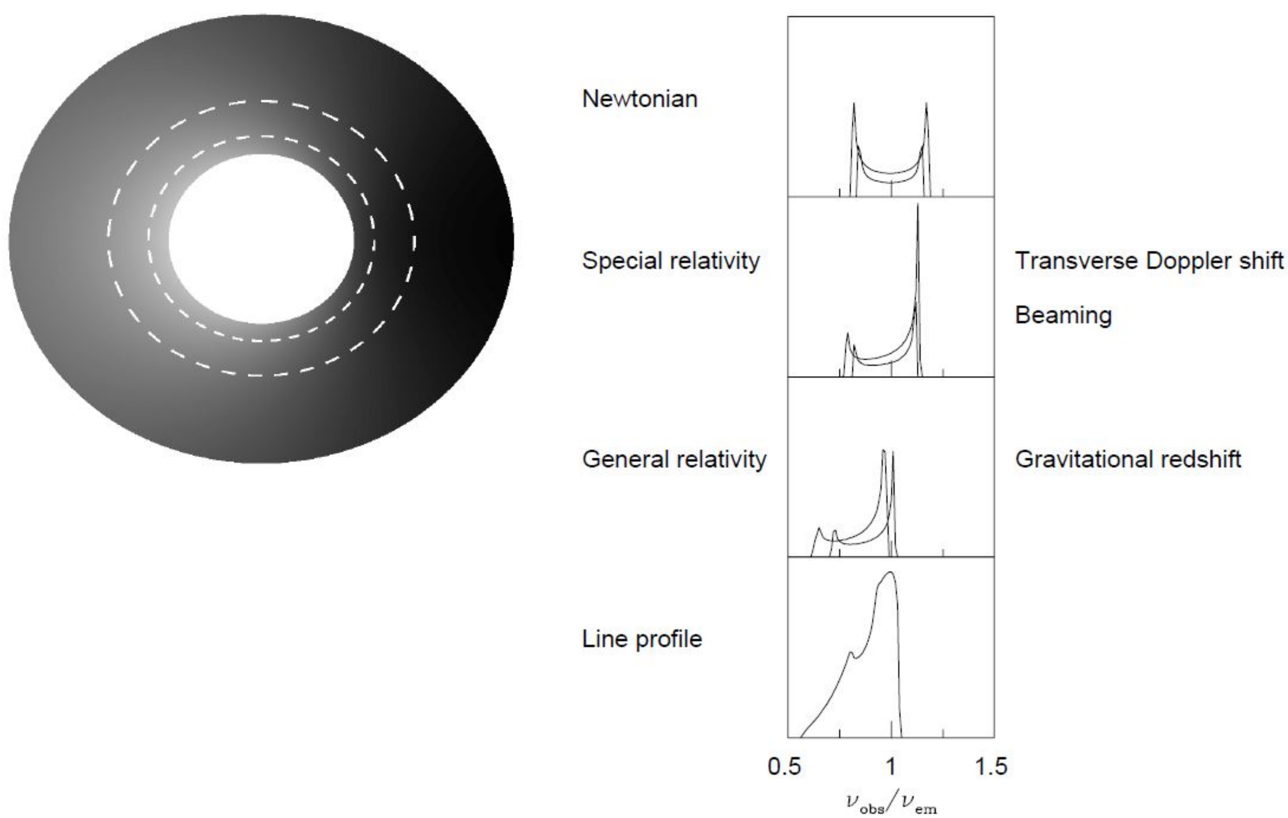


Figure 1.1: The profile of the broad iron line, image credit to Fabian (2000) [1].

Soft Comptonisation (multiple inverse Compton scattering by hot thermal electrons) occurs giving a power law X-ray spectrum. Flares from active or flaring regions irradiate the accretion disk forming a "reflection" component in the spectrum.

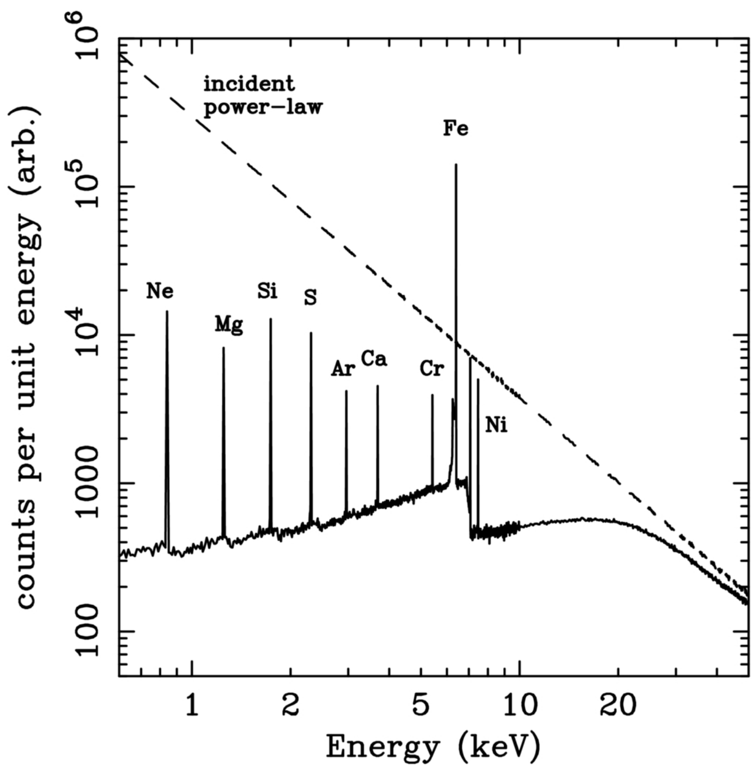


Figure 1.2: Reflection spectrum from a Monte Carlo simulation of an illuminated slab. The dashed line is the incident continuum and the solid line shows the reflected spectrum. Image credit Reynolds (1996) [1]

The iron line is produced when one of the two K-shell electrons is ejected following absorption of an X-ray, the threshold for neutral iron is 7.1 keV. The resultant state can decay through the dropping of an L-shell electron into the K-shell, releasing 6.4 keV as either an emission line photon (34%) or an Auger electron (66%). Ionised iron has a photoelectric threshold and the Kα of higher energy. The strength of the iron line is measured in terms of its equivalent width.

Equivalent Width

The equivalent width of a spectral line is the width of the rectangle which has the same area as that between the line and the continuum level.

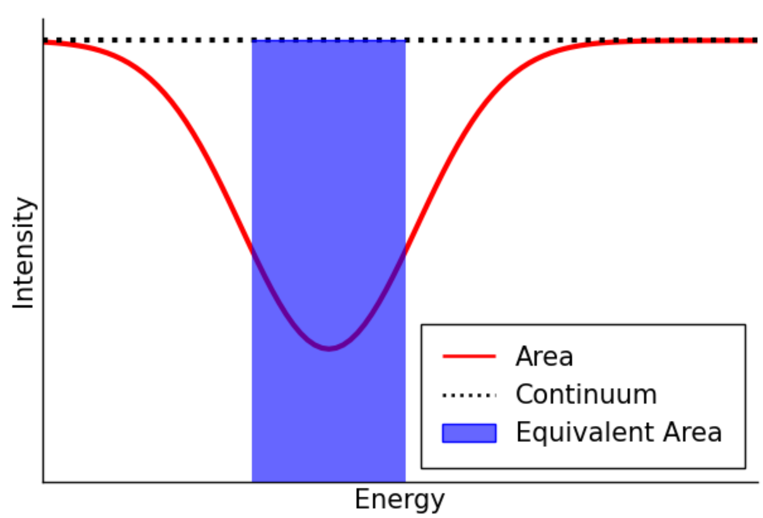


Figure 1.3: A diagram indicating the equivalent width. The width of the rectangle is the equivalent width.

The equivalent width is a function of the geometry of the accretion disk, the elemental abundances, the inclination angle, and the ionisation state.

! Key Points

- **X-ray Continuum**: The hard X-ray continuum in AGNs and GBHCs is thought to be from active/flaring regions in the corona. Thermal electrons multiply inverse Compton scatter optical and UV photons from the disk to X-ray energies. A hard X-ray power law results and irradiates the accretion disk to produce a reflection component.
- **Reflection Component**: The reflection component causes the spectrum to flatten above 10 keV as Compton recoil reduces the backscattered flux and results in a strong fluorescent iron line at ~ 6.4 keV. The energy and strength of this line depends on iron abundance, inclination of the disk, and its ionisation state. A strong iron absorption edge may also be present from an ionised disk.
- **Iron Line Profile**: Doppler shifts, relativistic broadening and gravitational redshifting give the broad and skewed line profile. As the line comes from the innermost regions, these effects are very pronounced and may be used to determine the black hole spin and inclination angle of the accretion disk. In most Seyfert 1 galaxies, the disk is inclined at around $30°$, consistent with the standard model.
- **Observations**: The iron line was first clearly detected by Ginga and a line profile subsequently resolved by ASCA confirmed the broad and skewed shape expected from an accretion disk around a Schwarzschild black hole. RXTE has been used to study the line and continuum variability on shorter timescales, with reduced energy resolution.
- **Black Hole Spin**: The radius of the ISCO decreases with spin. Since the line profile is sensitive to the innermost radius of fluorescent emission this may be used to estimate the black hole, with some astrophysical assumptions. With present time-averaged observations such measurements may be ambiguous as alternative models with different values for the spin may produce identical line profiles.
- **Alternative Models for the Production of the Broad Iron Line**: Alternative models that do not require an accretion disk appear to fail, in particular the line width cannot be entirely due to Comptonisation. Hybrid models in which the Comptonisation and Doppler/gravitational effects produce the profile are heavily constrained.
- **Variability**: Rapid X-ray continuum variability is observed in most AGNs, and the iron line is expected to vary in response to this with short time lag. Whilst these timescales are too short to be probed presently, significant and complex iron line variability has been observed. Often the line flux is seen to remain constant whilst the continuum changes, and there appears to be an anticorrelation between the reflected fraction and the equivalent width. In another study the reflected fraction and the photon index of the power law are correlated for individual objects and between different ones. These need to be explained as they appear contrary to our simple model of reflection. Flux correlated changes in the ionisation state of the disk may explain some of these facts.
- **Reverberation Mapping**: The rapid variability is associated with activation of new flares in the corona above the accretion disk. X-ray reverberation mapping uses observations of the iron line response to sudden changes in the continuum to study the accretion disk and black hole. This may be used to determine the geometry of the X-ray emission and the black hole spin and mass.
- **Other Classes of Object** Iron lines are observed in quasars, LLAGNs, radio-loud AGNs, and GBHCs. Quasars have the iron line decrease with increasing luminosity which may be due to the more luminous sources accreting closer to the Eddington limit being more highly ionised. Observations in LLAGNs may determine whether the accretion with low rates is an advection-dominated accretion flow or a thin disk. Weak lines have been observed suggesting that their low accretion rate flows are thin disks rather than optically thick ADAFs. THe broad iron lines and reflection humps are weak or absent perhaps because the reflection signature is swamped by a beamed continuum. GBHCs show smeared absorption and emission features in the reflection spectra, and there is a debate as to the precise nature of the accretion flow within a few tens of Schwarzschild radii.

2 Initial Gradus Testing

The below details initial experimentation with `Gradus.jl`. The code block below renders an image of a black hole with a thick accretion disk. Colour denotes the redshift of a given region on the disk.

```
using Gradus
using Plots

# Function defining the cross section of the disk
function discCrossSection(x)
    centre = 8
    radius = 3

    if (x < centre - radius) || (radius + centre < x)
        zero(x)
    else
        r = x - centre
        0.5(sqrt(radius^2 - r^2) + (0.5sin(3x)))
    end
end

# Metric Parameters
M = 1.0     # Mass
a = 0.7     # Black hole spin
 13 = 0.0
 22 = 0.0
 52 = 0.0
 3 = 0.0
i = deg2rad(70) # Inclination

# Instantiating the metric
m = JohannsenMetric(M, a,  13,  22,  52,  3)

# Position of the observer
# (x[1], Distance, Inclination, x[4])
x = SVector(0.0, 1000.0, i, 0.0)

# Maximum affine time, parameterises the solution
#  _max ~ 2*x[2]
 _max = 2000.0

# Setting up impact parameters
  = range(-10.0, 10.0, 100)
  = range(-10.0, 10.0, 100)
```

```
# Getting an array of velocities from the impact parameters
vs = vec([map_impact_parameters(m, x, a, b) for a in  , b in  ])
xs = fill(x, size(vs))

# Generating solutions
sols = tracegeodesics(m, xs, vs,  _max, save_on = false)

# Generating points and reshaping to be the same shape as the image
points = unpack_solution.(sols.u)
points = reshape(points, (100, 100))

# Redshift point function
redshift = ConstPointFunctions.redshift(m, x)
redshiftGeometry = redshift   ConstPointFunctions.filter_intersected()

# Disc
d = ThickDisc(r -> discCrossSection(r))

# this function returns the impact parameter axes
 ,  , image = rendergeodesics(
    m, x, d,  _max, pf = redshiftGeometry,
    # image parameters
    image_width = 200, image_height = 150,
    # the "zoom" -- use the impact parameter axes
     lims = (-20, 20),  lims = (-15, 15), verbose = false
)

heatmap( ,  , image, aspect_ratio = 1)
```

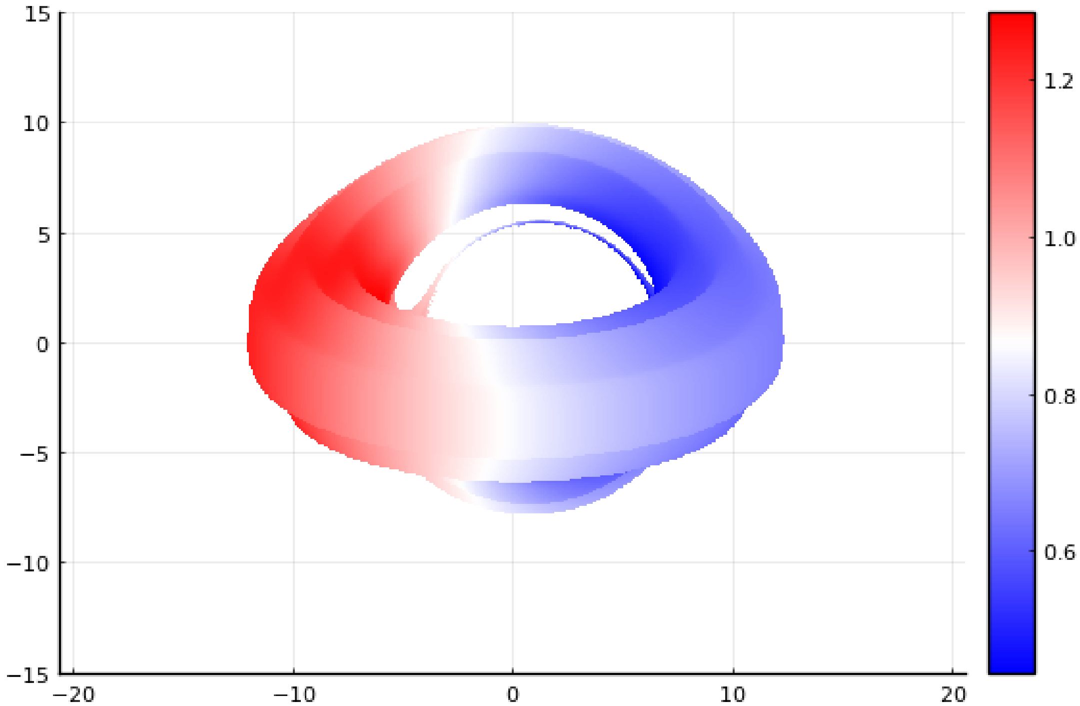


Figure 2.1: The first render produced.

Once familiarity was gained with the basics of `Gradus.jl` the line profile functionality was utilised through the following function.

```
function computeLineProfile(m, i, height, bins = range(0.0, 1.5, 180))
    # Position of the observer
    # (x[1], Distance, Inclination, x[4])
    x = SVector(0.0, 10000.0, i, 0.0)

    # Disk
    d = ThinDisc(0.0, Inf)

    # Setting up the model and profile
    model = LampPostModel(h = height)
    profile = emissivity_profile(m, d, model)

    # Computing the line profile
    bins, flux = lineprofile(m, x, d, profile; bins=bins, verbose=true)

    return bins, flux
end
```

This will allow for the investigation of the free parameters for the Johannsen metric; α_{13}, α_{22}, α_{52}, and ϵ_3. A lamppost corona model is used with variable height `h`. An example of the effect of varying the height of this corona is shown below.

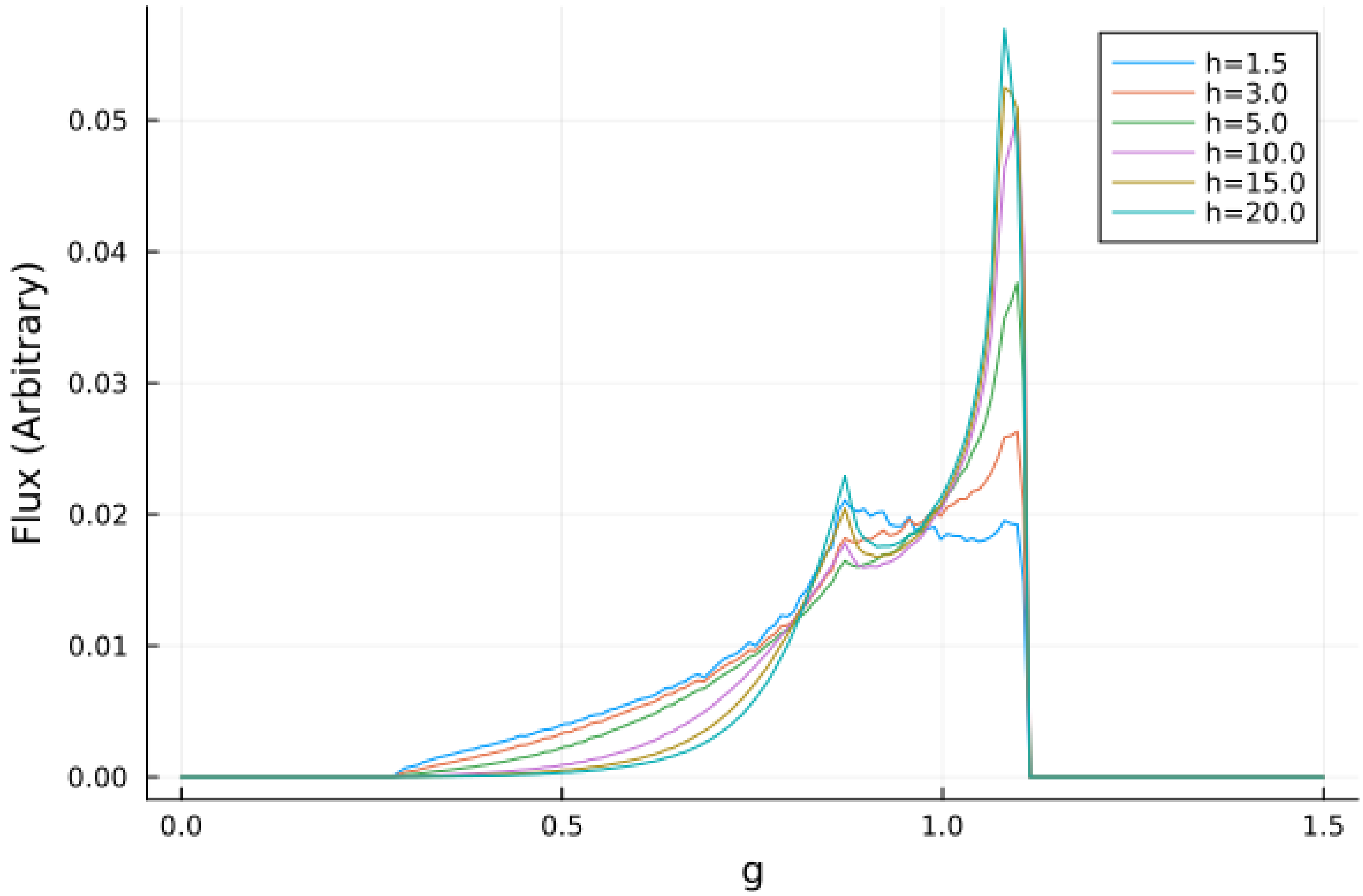
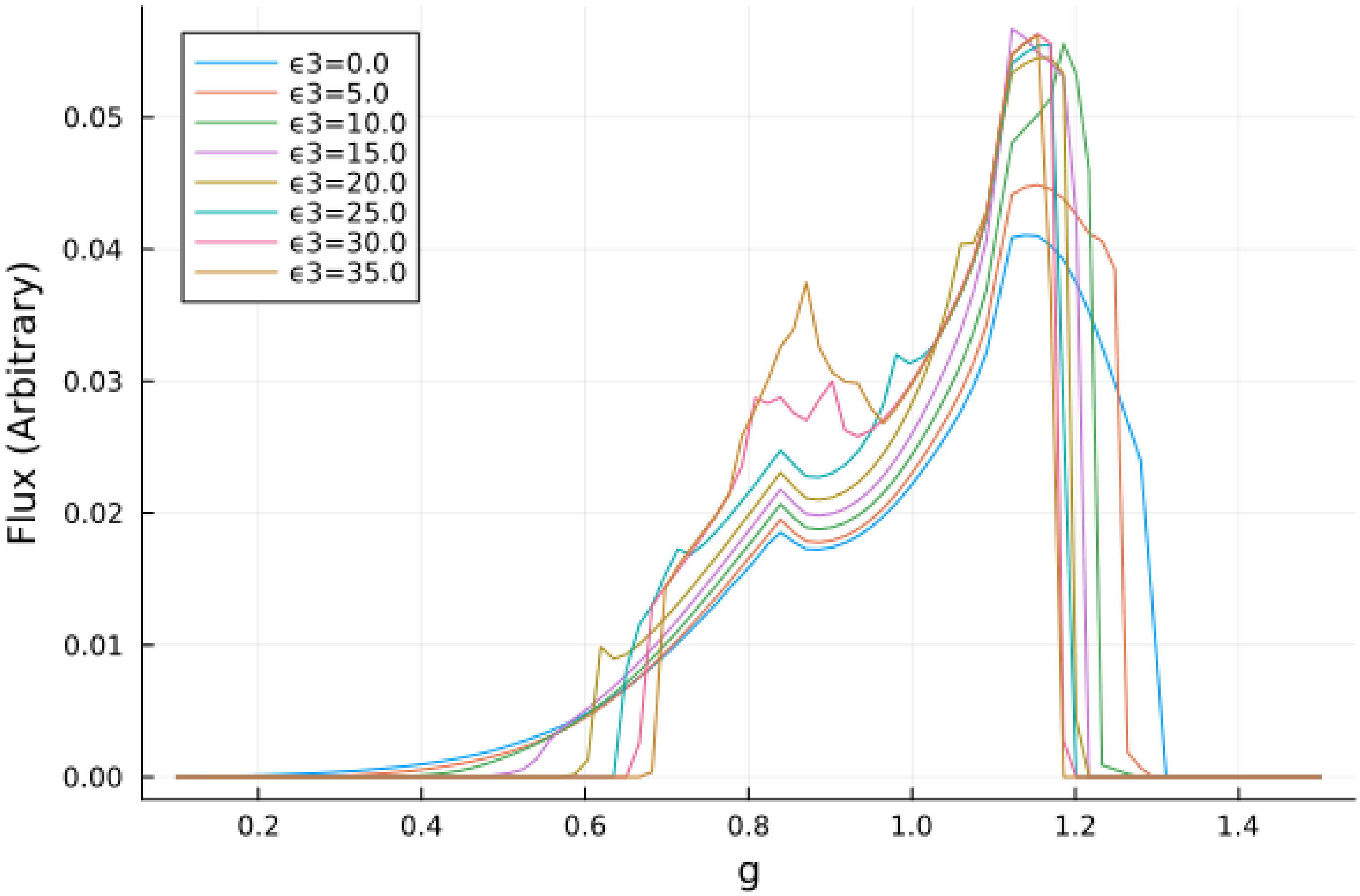


Figure 2.2: Initial variations of the height of the lamppost corona.

Following initial experimentation the two functions above were integrated together for efficient investigation. The two free parameters that will be the focus of this investigation are α_{13} and ϵ_3. Examples of the effects of these parameters on the emission lines are shown below


ϵ3=0.0
ϵ3=5.0
ϵ3=10.0
ϵ3=15.0
ϵ3=20.0
ϵ3=25.0
ϵ3=30.0
ϵ3=35.0
Flux (Arbitrary)
0.00
0.01
0.02
0.03
0.04
0.05
0.2
0.4
0.6
0.8
1.0
1.2
1.4
g

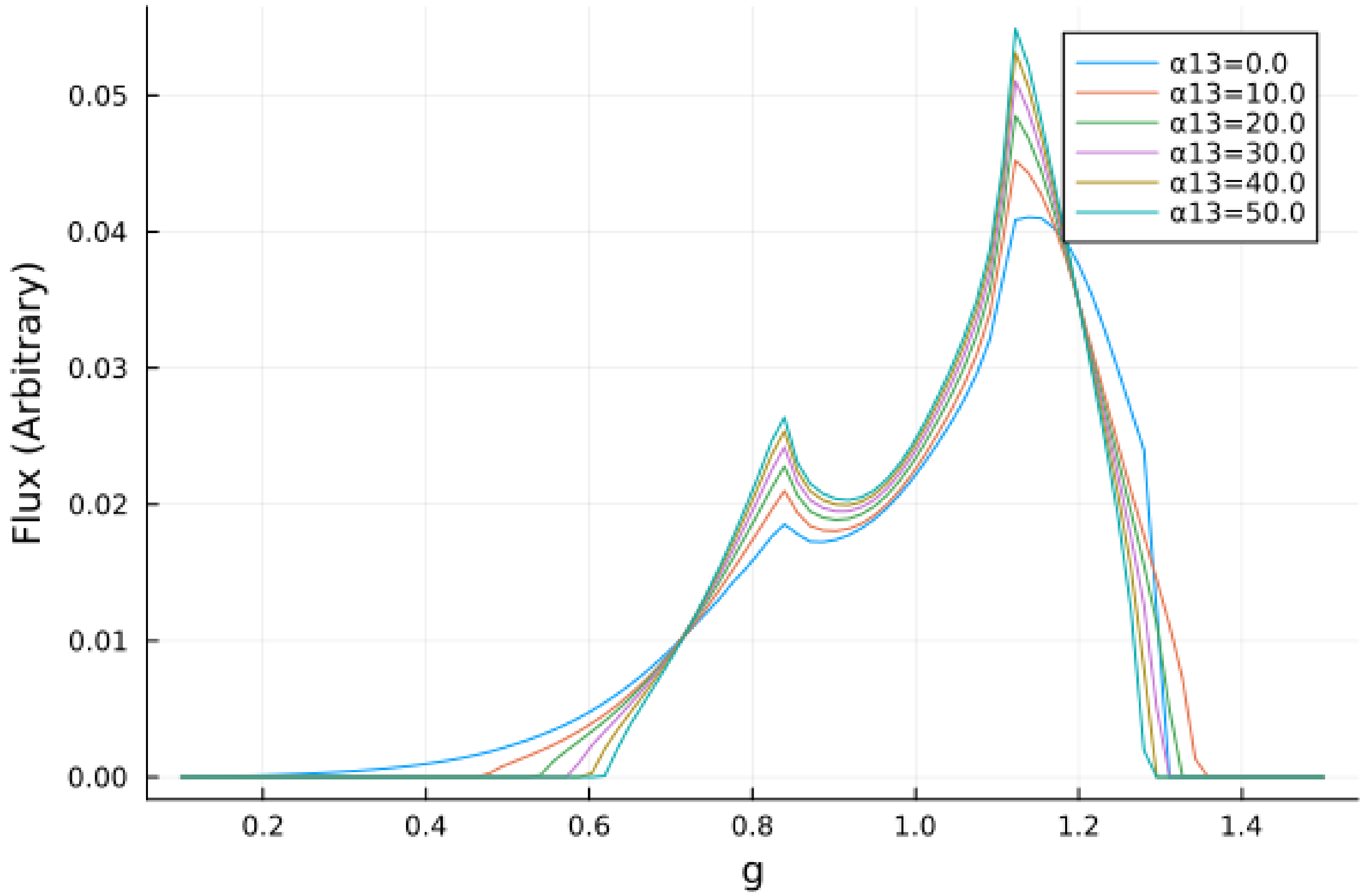


Figure 2.3: Initial variations of α_{13} and ϵ_3

It can be seen that α_{13} has a very smooth, almost linear effect on the shape of the line, while ϵ_3 is much more chaotic, particularly for $\epsilon_3 > 25$.

3 Parameter Space Investigations

Code was set up to easily run several variations to the model which will allow for the investigation of the parameter space. A thin disk accretion model and a lamppost corona were used. It was set up such that for a given configuration of h, a, i, ϵ_3, and α_{13} the line profile would be calculated and an image of the accretion disk produced and plotted together.

3.1 Ruling out α_{22} and α_{52}

Before exploring the 5D parameter space, tests were ran on both α_{22} and α_{52} such that their effects could be noted, should they be useful in the future.

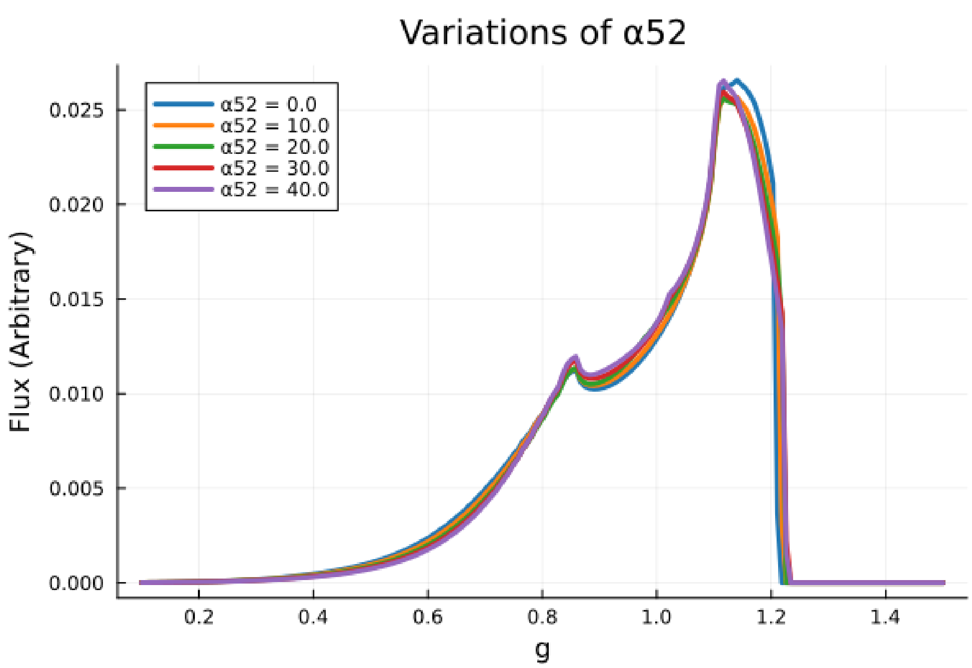


Figure 3.1: The resultant line profile from varying α_{52} between 0 and 40. A value of 50 did not successfully compute suggesting an upper limit on this parameter.

Varying α_{52} showed near random, and near zero effect on the line and will thus be disregarded for this investigation. Varying α_{22} often failed to compute and will thus similarly be disregarded for this investigation.

3.2 Exploring the 5-dimensional parameter space

Before varying the parameters it was necessary to determine the limits for useful variations. The spin is fundamentally limited to be $|a| < |M|$ and will thus be varied between 0 and 1. The ranges and default values for the parameter space are tabulated in Table 3.1.

Table 3.1: The ranges and default values for the parameter space.

Parameter	Minimum	Maximum	Default
α_{13}	0.0	50.0	0.0
ϵ_3	0.0	30.0	0.0
a	0.0	1.0	0.998
h	R_{ISCO}	30 R_G	10 R_G
i	0.0°	90.0°	60°

3.2.1 Line profiles

- Figure 3.2 shows that increasing the free parameter α_{13} sharpens the two peaks of the line, increases the maximum flux, and makes it more narrow.
- Figure 3.3 shows that increasing the free parameter ϵ_3 similarly increases the maximum flux, however the resultant profile is much more noisy. While α_{13} provided sharp peaks, ϵ_3 provided broader peaks, particularly in the shorter one.
- Figure 3.4 shows that increasing the inclination smoothes out the line profile and shifts the peak to higher energies. The line is broadened significantly.
- Figure 3.5 shows that increasing the corona height makes the line profile sharper, increases the maximum flux, and shifts the maximum to slightly lower energies.
- Figure 3.6 shows that the spin had the least prominent impact on the profile of the line, seeming to only slightly decrease the height of the peaks with increasing spin. This may be an effect of the small range of spins available. It may therefore be informative to investigate this further by increasing the mass, using different corona heights, or varying the inclination.

3.2.2 Redshift Image Renders

Alongside the line profiles, images of the accretion disk were rendered with a colourmap denoting the redshift of light from a given part of the disk. The corona height was not varied here as it did not impact the render.

Some of the parameters appeared to change the radius of the ISCO. This was investigated by explicitly finding the ISCO, as tabulated below.

Parameter	Value	ISCO (R_G)
α_{13}	0.00	1.24
	16.67	8.62
	33.33	11.70
	50.00	14.03
ϵ_3	0.00	1.24
	10.00	2.36
	20.00	2.90
	30.00	3.51
a	0.00	6.00
	0.32	4.92
	0.63	3.68
	0.95	1.94

The ISCO is proportional to both α_{13} and ϵ_3, and inversely proportional to the spin.

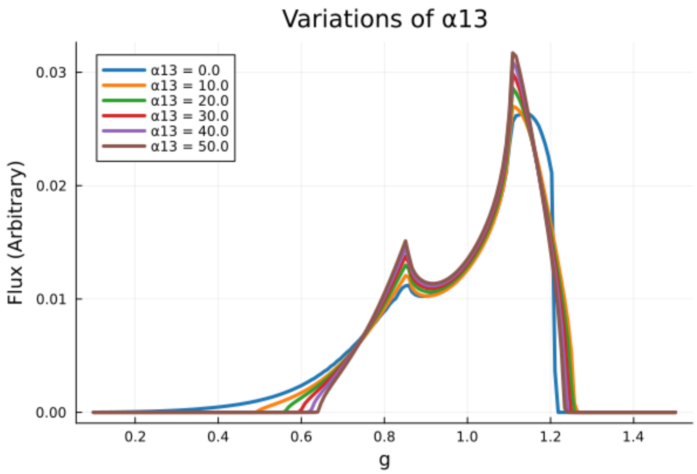


Figure 3.2: The line profiles resultant from varying α_{13} between 0 and 50.

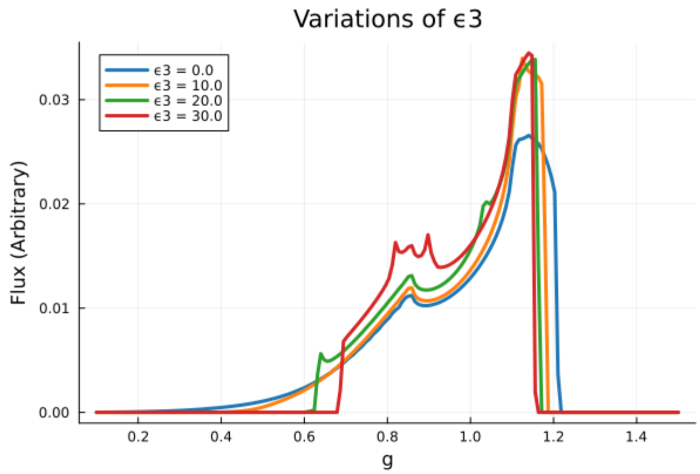


Figure 3.3: The line profiles resultant from varying ϵ_3 between 0 and 30. Values of 40 and 50 were tested however caused an error, there must be an upper bound on the allowed values of this parameter.

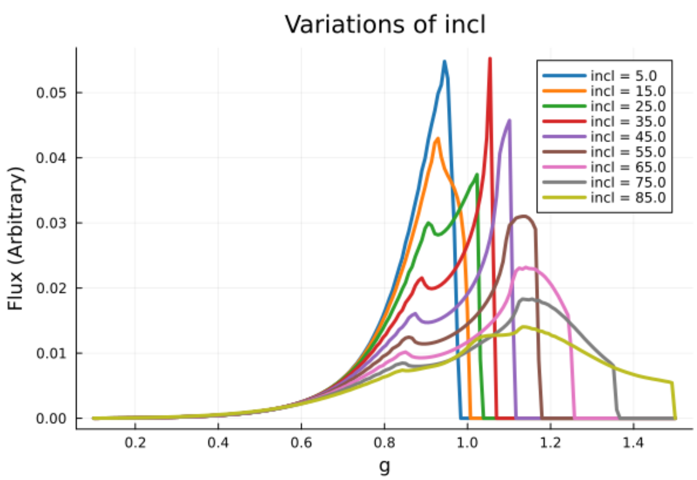


Figure 3.4: The line profiles resultant from varying the inclination between 5 and 85 degrees.

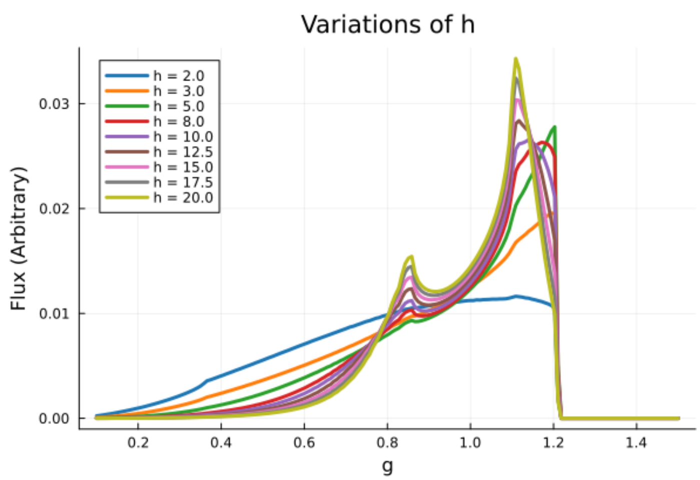


Figure 3.5: The line profiles resultant from varying h between 5 and 30 gravitational radii.

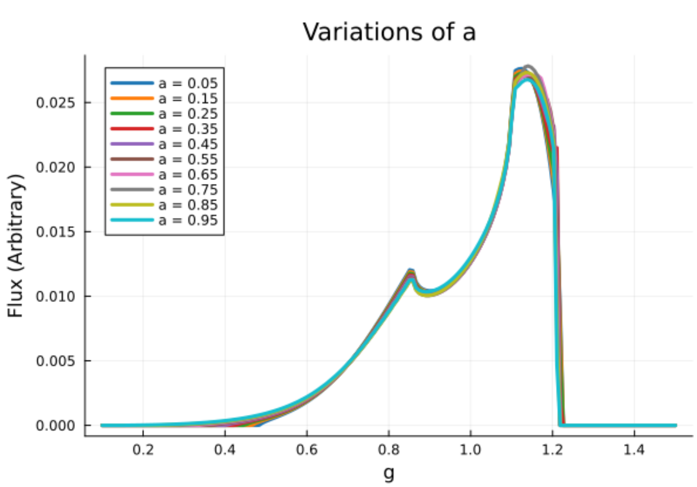


Figure 3.6: The line profiles resultant from varying a between 5 and 95% of $M (= 1)$.

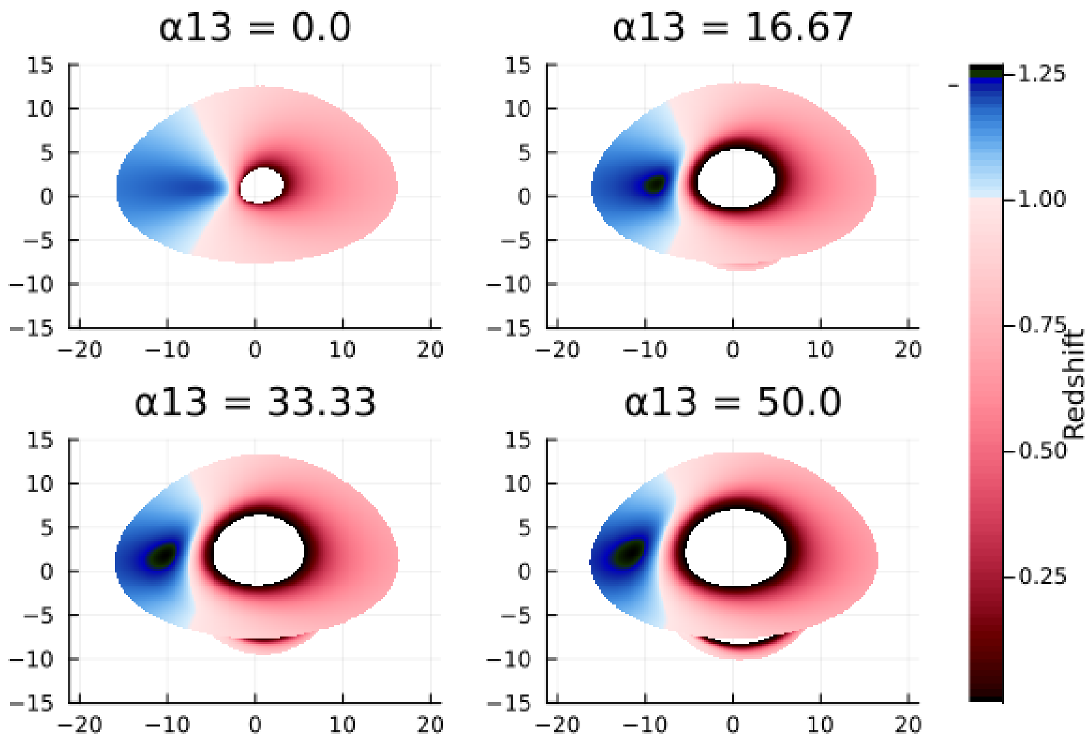


Figure 3.7: The effect of α_{13} on the accretion disk render. The ISCO appears to grow with α_{13}

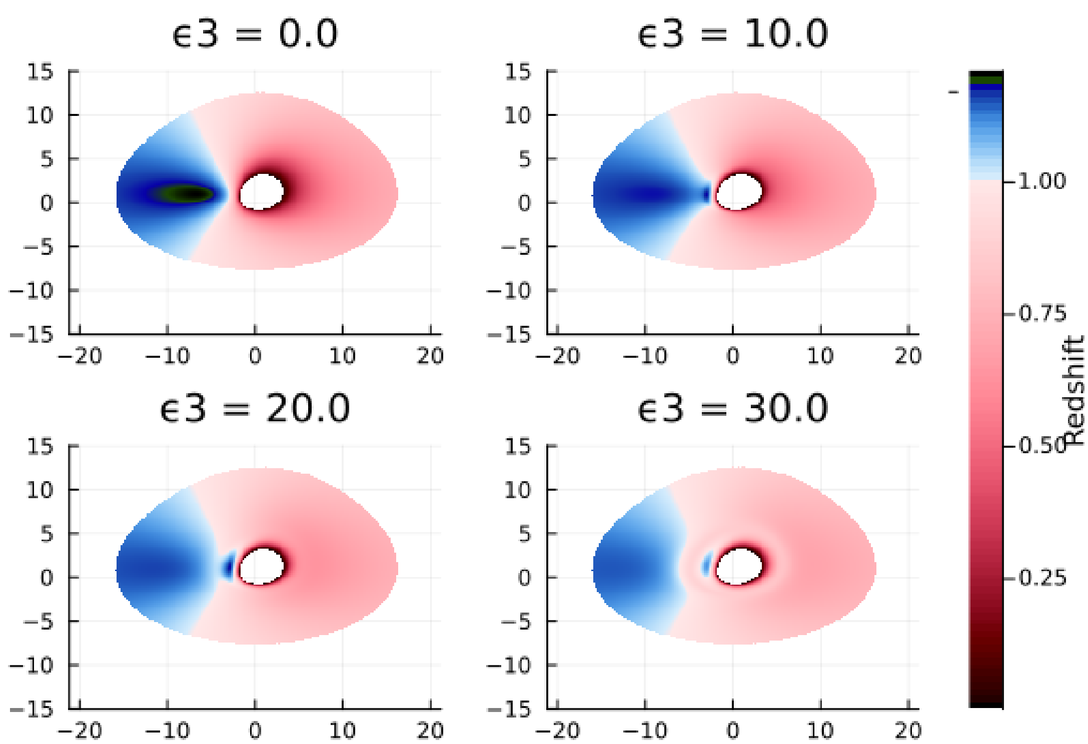


Figure 3.8: The effect of ϵ_3 on the redshift image. The shape of the disk does not change but the redshift profile does, particularly in the shrinking of the blueshift region, and an overall decrease in the degree of redshifting.

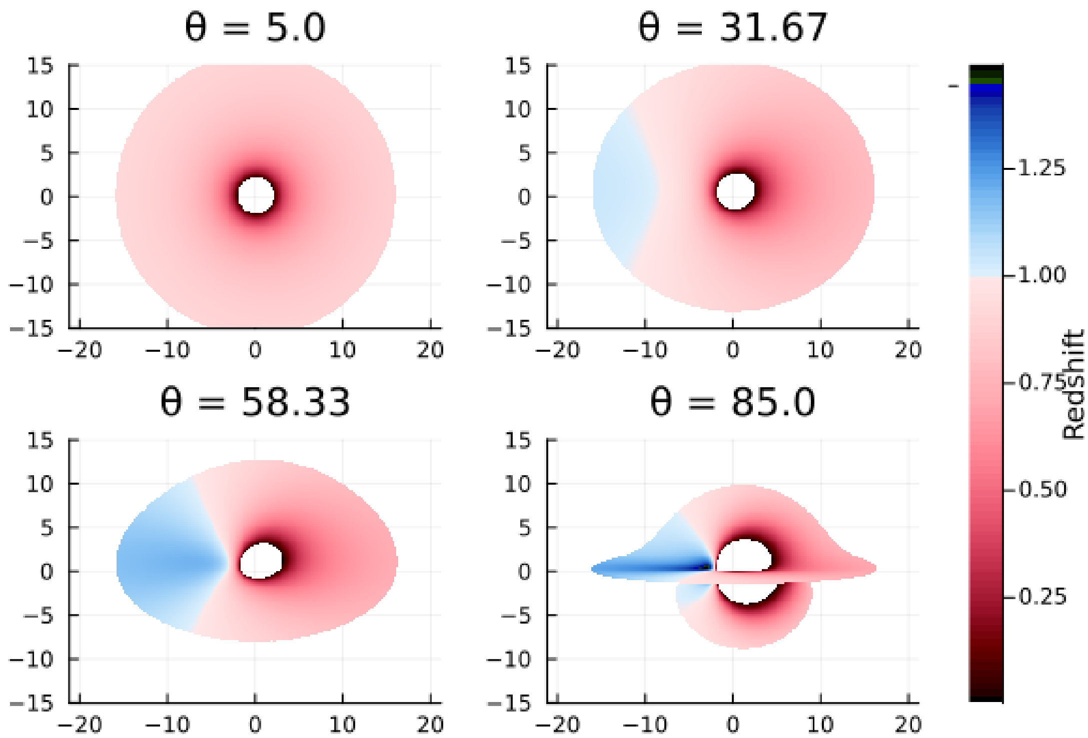


Figure 3.9: The effect of inclination on the redshift image. The redshift becomes more apparent as the inclination increases and the disk becomes edge on. This agrees with the observed line profile trend of energy increasing with inclination as the effects of doppler shifting become more apparent with increasing energy.

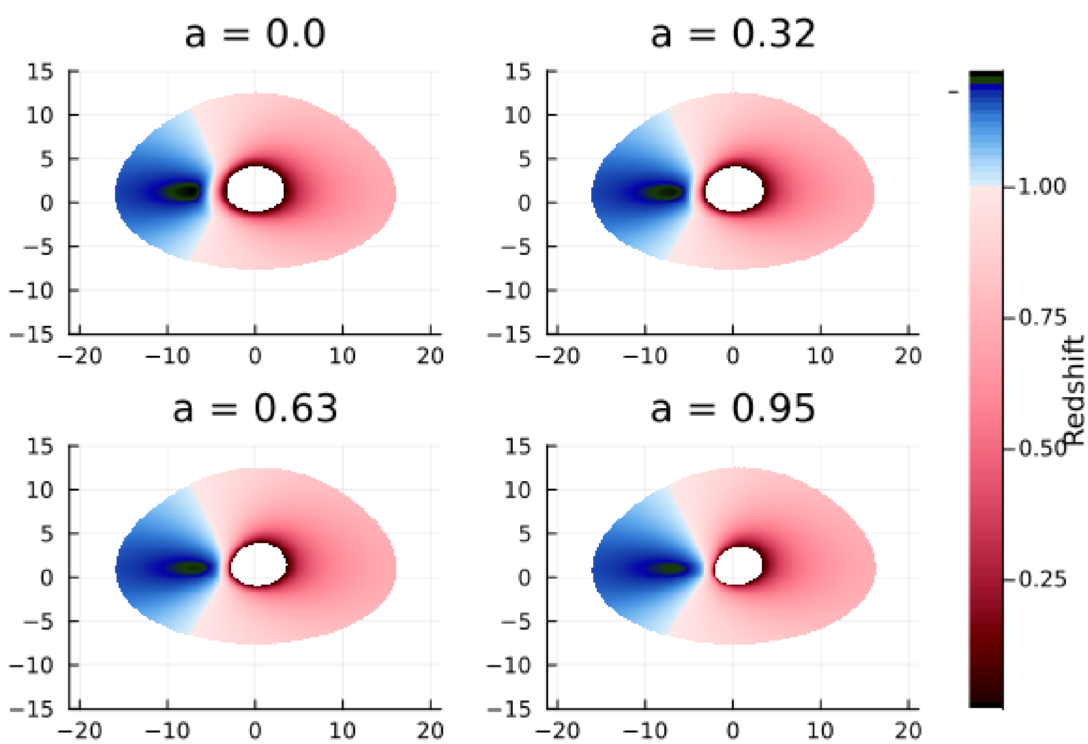


Figure 3.10: The effect of black hole spin on the redshift image. The ISCO appears to shrink.

3.3 Summary

> **!** Summary
>
> - α_{22} and α_{52} had unpredictable or minimal effect on the line profile and rendered images so are therefore to be disregarded.
> - a: Increasing the spin appears to shrink the ISCO of the accretion disk. Its effect on the line profile is less obvious, it appears to decrease the flux slightly but is otherwise negligible.
> - h: Increasing the corona height makes the line profile sharper and increases the peak flux, while decreasing the energy of the peak. It has no effect on the renders.
> - θ: Increasing the inclination (where 0 degrees has the accretion disk face-on) smoothes out the line profile and increases the peak energy. The effect of redshift becomes more apparent with increased inclination.
> - α_{13}: Increasing α_{13} appears to increase the radius of the ISCO and make the line profile sharper.
> - ϵ_3: Increasing ϵ_3 increases the height of the line profile, and makes it more chaotic.
> - The ISCO is proportional to both α_{13} and ϵ_3, and inversely proportional to the spin.

3.4 Code Specifics

The workhorse of the parameter investigations was the utility function `JohannsenParamVar` which sets up the metric and observer position and then calls either `ComputeLineProfile` or `RenderImage`. This utility function was made versatile using keyword arguments allowing for different methods to be called within it. This was then wrapped in `ParamLoop` which took the index (the variable to be looped through) and a set of values to call `JohannsenParamVar` for.

To maximise readablity of the redshift images, a custom colormap is made through `RedshiftColormap` which gives red for any values < 1, blue for those > 1, and white for those $= 1$. This gave the images seen in Figure 3.7, 3.8, 3.9, 3.10.

4 Fitting Spectral Lines

With the parameter space explored, it was useful to setup the architecture for fitting the model to spectral data. This was done using `SpectralFitting.jl` and defining a custom model using Gradus which fits the following parameters:

Parameter	Meaning
`K`	Normalisation
`h`	Corona Height
`E`	Line Energy
`R_in`	Inner Radius
`R_out`	Outer Radius
	Inclination
`a`	Spin
`13`	α_{13} Parameter
`3`	ϵ_3 Parameter

The architecture of this model was heavily based upon previous work on GitHub.

The model is instantiated using the `LampPostJohannsen` utility function. When called, the parameters are used to instantiate a `JohannsenMetric`, observer position, `ThinDisc`, and an emissivity profile. These are then used to compute the line profile as previously done which is returned.

All fitting is handled by `SpectralFitting` through the Levenberg Marquadt algorithm.

To test the implementation a line profile was computed and random noise added to it. This was then passed through the fitting function. The fitting process currently takes a long time and yields poor results so more work is needed. Time was taken to streamline the functionality of main2.jl via keyword arguments and general code cleanup.

Upon fitting, the error `Warning: No parameter uncertainty estimation due to error: LinearAlgebra.Singular` occurs and therefore the fit is not currently quantified.

With the current errors present it was decided to attempt the fit on actual data in order to determine whether they are a result of the data, or the fitting algorithm.

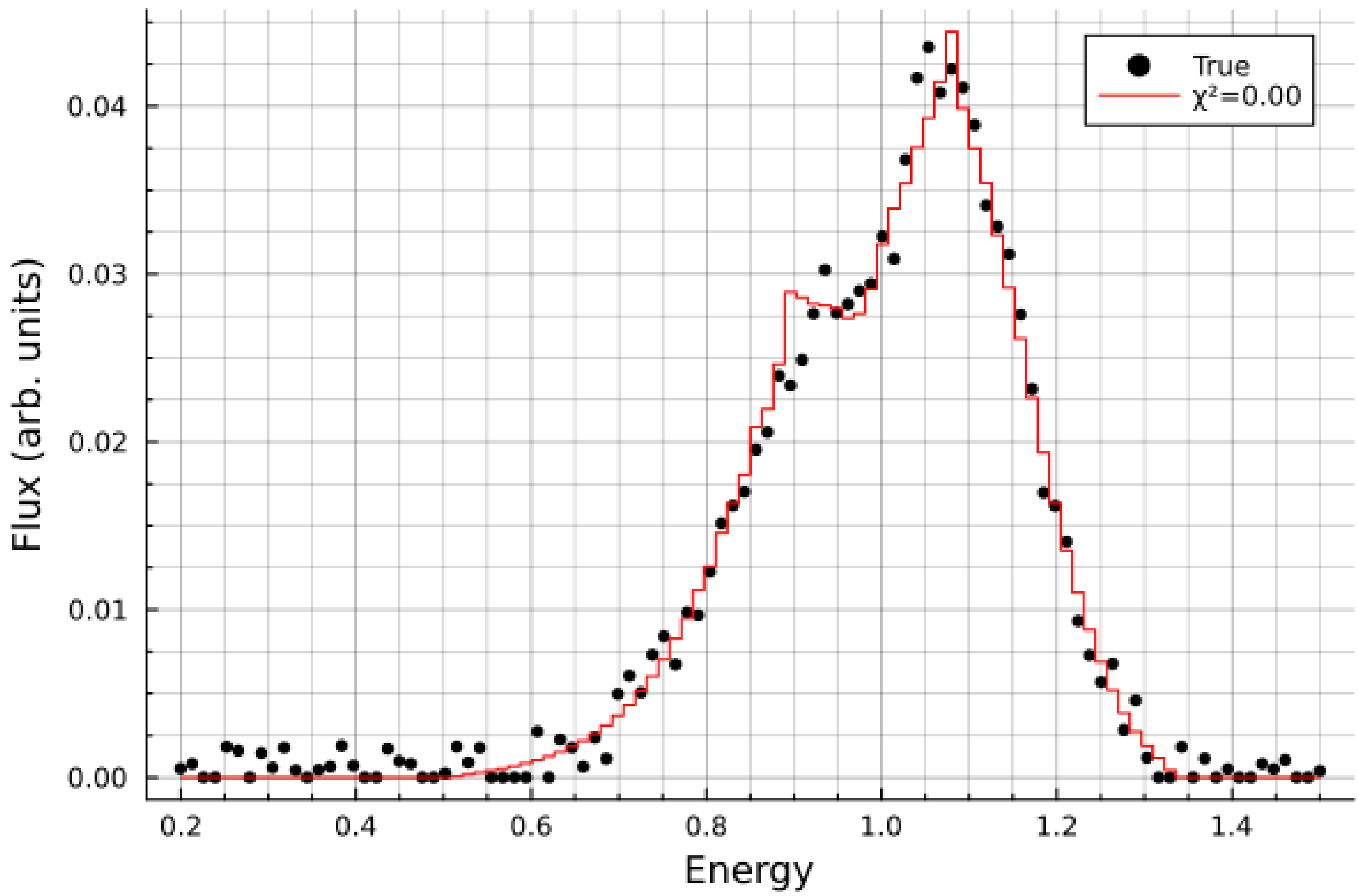


Figure 4.1: The first attempt at fitting to the noisy simulated data. This fit took 67 iterations and 15.3 minutes to complete.

5 Fitting to Data

Data from the NuSTAR (Nuclear Spectroscopic Telescope Array) spacecraft [2] was used, and the dataset used for testing was `nu80402315002B01`. Data loading was handled by `SpectralFitting.jl` through the `OGIPDataset` struct. This takes the spectrum file as an argument and automatically pulls in the background, `.arf`, and `.rmf` files to compute the spectrum.

5.1 Initial Results

Table 5.1: The ranges and default values for the parameter space.

Model	Variable	Value	Error	χ^2
Johannsen	a	0.57	1.88	116.84
	h	50.00	8.77	
	θ	58.24	4.93	
	α_{13}	2.29	10.96	
	ϵ_3	29.74	25.89	
	K	161.8	7.15	
	E	6.27	0.016	
Kerr	a	0.77	0.057	124.64
	h	50.0	2.75	
	θ	55.44	2.67	
	K	162.0	7.07	
	E	6.27	0.014	

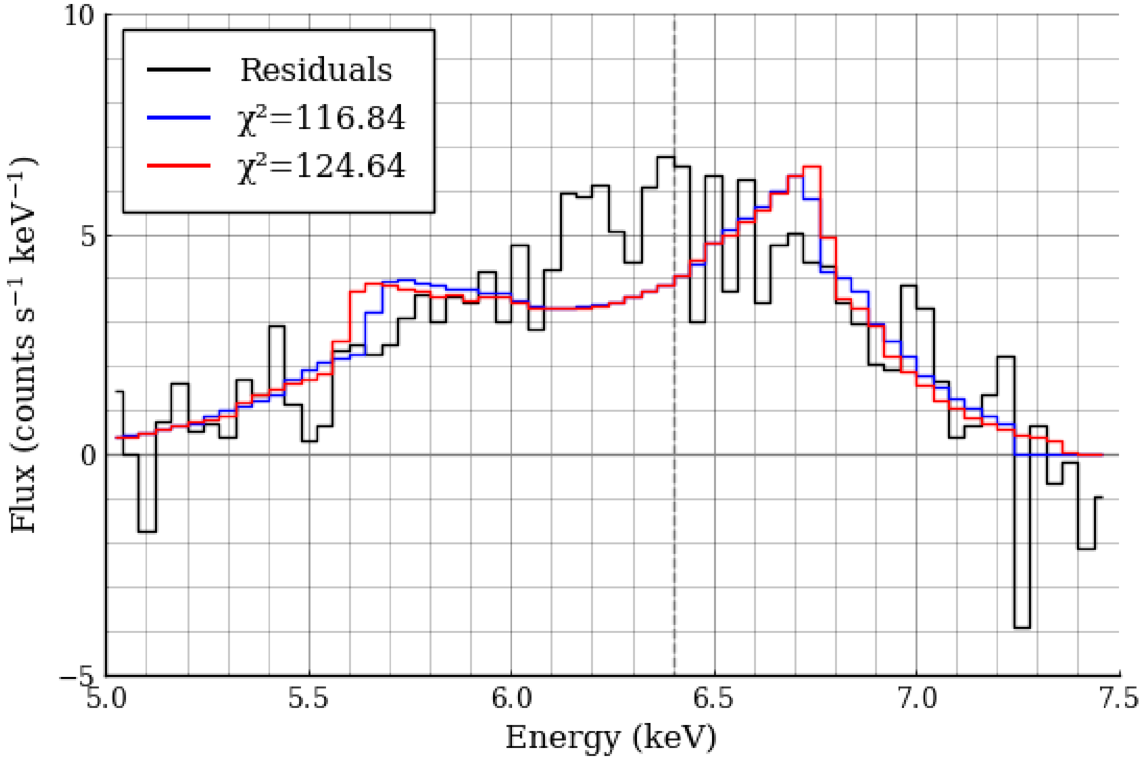


Figure 5.1: The first attempt at fitting to data. The Johannsen metric was used to produce the blue line, and the Kerr metric was used to produce the red. Neither fit was particularly good which may either be the result of poor data, or a poor model.

The NuSTAR datasets have two distinct spectra from each telescope, which can be fit simultaneously, binding some of the parameters together. The fitting code was thus adjusted to fit both datasets simultaneously while binding all parameters other than the normalisation. The dataset labelled "nu80402315002" was once again used.

It is evident from Figure 5.2 that the fitting process needs refinement as the fits do not currently fit the data well, and therefore drawing comparisons would be useless.

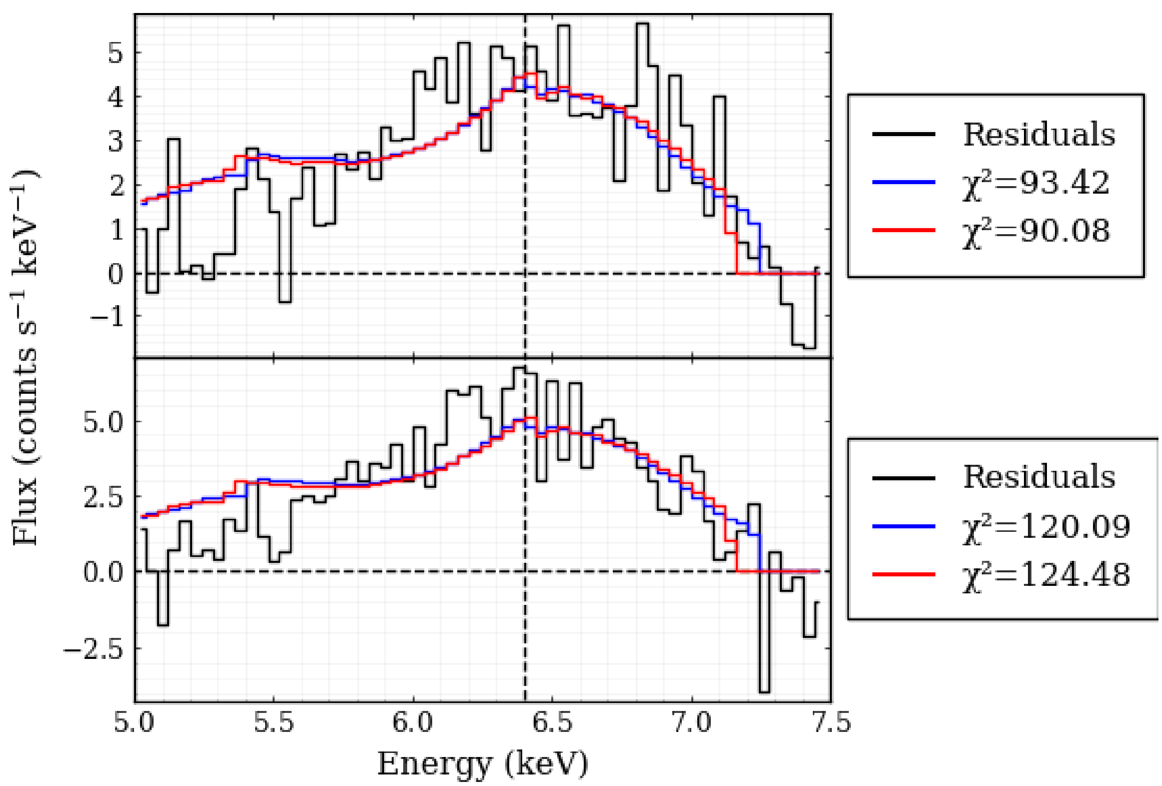


Figure 5.2: The fits to the nu80402315002 dataset. The Johannsen fit is shown in blue, and the Kerr fit in red.

References

[1] A. C. Fabian *et al.*, Publications of the Astronomical Society of the Pacific **112**, 1145 (2000).

[2] F. A. Harrison *et al.*, American Physical Journal **770**, 103 (2013).